MODULE 01 · ALL DOMAINS

The Claude Developer

What the exam tests, the developer platform on one page, ten rules behind the right answers, and the company we build for.

DOMAINS All 8 ITEMS 53 in 120 min MODULE 1 of 36

admin@sthithapragna.com

The exam rarely asks how to call Claude. It asks whether what you built would **survive production**.

Who the exam is for

- **Engineers who build with Claude:** the Claude API, Claude Code and MCP, from first integration to production agents.
- **Typical candidate:** 1-5 years of software engineering, 6+ months hands-on with Claude or a similar LLM system, Python or TypeScript.
- **Assumed knowledge:** REST APIs, CLI tools, LLM fundamentals, agents, context management and MCP.
- **Out of scope:** casual or non-technical users, people without hands-on development, prompt-writing-only roles.

> ON THE EXAM
>
> The guide recommends building and running at least one Claude application before you sit the exam.

The exam at a glance

ITEM	DETAIL
Questions	53, multiple choice and multiple response
Time	120 minutes, about 2 min 15 s per question
Scoring	Scaled 100 to 1,000; pass at 720
Result	Pass or fail, plus percent correct by domain
Delivery	Pearson VUE, online or at a test centre
Fee and validity	USD 125; valid for 12 months

ON THE EXAM

Every question states how many answers to choose. Read that number before you read the options.

Eight domains, eight weights

DOMAIN	WEIGHT	MODULES
D1 Agents and workflows	14.7%	23 to 26
D2 Applications, integration	33.1%	03, 09 to 17
D3 Claude Code	3.1%	27
D4 Testing and debugging	2.6%	30
D5 Models, optimisation	16.8%	02, 04, 06, 18, 19
D6 Prompts and context	11.0%	05, 07, 08
D7 Security and safety	8.1%	28, 29
D8 Tools and MCPs	10.6%	20 to 22

ON THE EXAM

The course gives each domain teaching time in proportion to its published weight.

Where the marks are

- **Domain 2 alone is a third of the exam.** API mechanics, application design, engineering basics, configuration.
- **Domains 2 and 5 together are half.** Fundamentals and model choices sit under almost everything.
- **Agents, prompts and tools** add up to about 36%.
- **Claude Code and debugging are 5.7% combined.** Learn them well, but do not over-revise them.

> ON THE EXAM
>
> The practice bank we analysed gave Domain 2 about a quarter of its items. This course follows the guide, not the bank.

The developer platform on one page

Agent SDK
Claude Code's agent loop
Runs in your process

Messages API
One stateless request
Streaming, Batch, vision
The base of everything

Managed Agents
Anthropic-hosted loop
Beta

Tools
Your functions (client)
Web search, code (server)

Claude models
1M-token context
Thinking on by default
Priced per token
Opus, Sonnet and Haiku tiers, with Fable above Opus

MCP
Standard server protocol
Tools, resources, prompts

Claude Code
Agentic coding tool
CLAUDE.md, settings, hooks

Skills
Packaged instructions
Loaded on demand

Platforms
Claude API, Bedrock
Google Cloud, Foundry, AWS

THE TAKEAWAY

Most questions are about choosing between these pieces. Know what each one is for.

What the Messages API gives you

FEATURE	REACH FOR IT WHEN
Streaming	Users watch the answer arrive, or outputs are long
Message Batches	Nobody is waiting: half price, within 24 hours
Vision and PDFs	The input is a photo, scan or document
Thinking and effort	The task needs reasoning; trade speed for depth
Prompt caching	A long prefix repeats across calls
Structured outputs	Code, not a person, reads the answer
Files and citations	Reuse uploads; tie answers to sources
Tool use	Claude must fetch live data or take an action

ON THE EXAM

Each feature answers one kind of need. Most scenario questions are a need looking for its feature.

Who runs the agent loop?

OPTION	WHO RUNS THE LOOP	CHOOSE IT WHEN
Messages API	You write it	The loop is simple
Claude Agent SDK	The SDK, in your process	You want agent features
Managed Agents	Anthropic's infrastructure	You want it hosted

THE TIEBREAKER

The loop is the same idea in all three. The difference is who maintains it and where it runs.

Where Claude runs

PLATFORM	WHAT TO KNOW
Claude API	Run by Anthropic; new features arrive here first
Claude Platform on AWS	Run by Anthropic; AWS sign-in and billing
Amazon Bedrock	Run by AWS; regional; fewer API features
Google Cloud's Agent Platform	Formerly Vertex AI; regional endpoints
Microsoft Foundry	Hosted on Azure or on Anthropic

ON THE EXAM

When a contract names a platform, the platform is a requirement, not a preference.

The smallest useful request

```python
import anthropic
client = anthropic.Anthropic()  # reads ANTHROPIC_API_KEY
message = client.messages.create(
    model="claude-sonnet-5-5",
    max_tokens=1024,
    system="You answer tenants' questions about their tenancy.",
    messages=[{"role": "user", "content": "When is my rent due?"}],
)
for block in message.content:
    if block.type == "text":
        print(block.text)
```

4 From Claude 4.6 on, a model ID is a fixed snapshot.

5 Required. A hard cap on output, thinking included.

9-11 Content is a list of blocks. Do not assume the first is text.

THE TAKEAWAY

The exam asks what each part decides, not how to type it.

Ten rules explain almost every right answer on this exam.

The ten rules: one to five

RULE	WHAT IT MEANS
1. Code enforces, prompts guide	A rule that must always hold gets a check outside the model
2. Evidence before change	Read traces, logs and usage before editing anything
3. Requirement first	Decide what matters, choose, then test on real inputs
4. The complete answer	The right option covers every need the question states
5. Use what the platform gives	SDKs, loops and features before custom builds

The ten rules: six to ten

RULE	WHAT IT MEANS
6. Change in small, tested steps	Regression tests, batches, impact checks, then roll out
7. Failure is a designed path	Retry, repair or fall back, then fail cleanly
8. Keep context lean	Prune spent output, isolate steps, reset stale sessions
9. Variation is a property	Manage it with validation; do not call it a bug
10. Least privilege	Separate keys, scoped tools, hooks where risk is

How to use the rules

- **As tiebreakers.** When two options would both work, a rule usually decides.
- **As a smell test.** An option that makes the prompt the only guard, or skips the evidence, is usually a distractor.
- **As a checklist.** If the question names three needs, the answer covers three.
- Module 33 drills the rules against the distractor patterns the exam uses.

> BE CLEAR
>
> The ten rules are this course's reading of the practice questions, checked against Anthropic's documentation. They are not official wording.

Meet Copperlet

- **A fictional UK lettings software company:** tenancies, rent, deposits, repairs and safety certificates for agents, landlords and tenants.
- **About 1,400 agencies** managing **310,000 tenancies**; 220 staff, 60 engineers in 8 squads, Manchester.
- **Python services, a TypeScript app, PostgreSQL, GitHub.** Separate development, staging and production.
- **Claude API for most features;** Claude Code for every engineer; one customer contract requires Amazon Bedrock.
- **Tenant data is personal data,** so security and privacy are never optional.

> YOUR ROLE
>
> You are a developer on Copperlet's six-person AI platform team.

Seven features, seven anchors

COPPERLET FEATURE	WHAT IT TEACHES	MAINLY IN
Tenant help assistant	Streaming, sessions, context	08, 10, 12
Repair-request triage	Workflows, structured output	07, 10, 23
Document reader	Vision, PDFs, citations	03, 07, 28
Overnight compliance sweep	Batch, rate limits, cost	12, 13, 19
Agency-ops agent	Agent SDK, tools, hooks	20, 24, 25
Property-data MCP server	Building and reusing MCP	21, 22
Claude Code rollout	CLAUDE.md, settings, CI	16, 27, 29

ON THE EXAM

Scenarios also use fresh companies, so you practise transferring each rule to a new setting.

Things that have moved since the guide

- **Guide v1.0 is dated July 2026.** The platform has kept moving since.
- **The 5.5 models** (Sep-Oct 2026): thinking on by default; forced tool choice rejected on the newest models.
- **Fast mode** generally available on Opus 5.5 since 15 Oct 2026.
- **MCP 2026-07-28:** stateless protocol; sampling, roots and logging deprecated.
- **Claude Code today:** auto mode is the starting mode; custom commands merged into Skills.

> THE RULE
>
> Answer with the engineering principle. The version detail is a currency check, and module 32 collects every one.

How this course works

- **Pause and commit.** Every scenario stops. Choose before the walkthrough starts.
- **Choose two.** Some scenarios want two answers, as some real questions do.
- **Code on screen.** Read it for shape and decisions. Nobody asks you to type it.
- **Three practice exams** at the end make one full 53-question sitting.

> ON THE EXAM
>
> Every scenario and practice question in this course is new. None comes from a question bank.

> Copperlet's AI platform team is building the agency-ops agent. It reads files, calls the payments service and the CRM, needs hooks to block risky actions, and keeps sessions between runs.
>
> The agent process must run inside Copperlet's own cloud account, next to the payments service. The team does not want to write or maintain its own agent loop.

Which approach fits best?

A. Call the Messages API directly and write the tool loop, history and session storage by hand

B. Use Claude Managed Agents so Anthropic hosts the agent loop and its sessions

C. Use the Claude Agent SDK inside a container in Copperlet's cloud account

D. Have an engineer run each investigation by hand in an interactive Claude Code session

◆ PAUSE AND DECIDE

Which approach fits best?

A Call the Messages API directly and write the tool loop, history and session storage by hand

Would work, but it is exactly the loop the team refused to maintain.

B Use Claude Managed Agents so Anthropic hosts the agent loop and its sessions

The harness runs in Anthropic's infrastructure. The process must run in Copperlet's account.

C Use the Claude Agent SDK inside a container in Copperlet's cloud account

Loop, tools, hooks and sessions come built in, in a process Copperlet operates.

D Have an engineer run each investigation by hand in an interactive Claude Code session

A person in a terminal is not a product feature.

THE SIGNAL

'Must run in our account' plus 'no loop to maintain' points to the Agent SDK, self-hosted.

An online retailer runs a nightly job that writes product descriptions with Claude. Last month the job's bill doubled, although the number of products was flat.

The engineering manager wants the cost back down this week. Nobody has yet looked at what changed.

What should the team do first?

A Move every request in the job to the smallest, cheapest model tier

B Turn on prompt caching for every request the job sends

C Halve the max tokens limit on every request in the job

D Break the job's usage down by model and by input, output and cache tokens to find what changed

◆ PAUSE AND DECIDE

What should the team do first?

A Move every request in the job to the smallest, cheapest model tier

May cut quality as well as cost. It changes something before knowing what broke.

B Turn on prompt caching for every request the job sends

Helps only if a long, repeated prefix exists. Unknown until usage is read.

C Halve the max tokens limit on every request in the job

Truncates output if the length was needed. A guess, not a diagnosis.

D Break the job's usage down by model and by input, output and cache tokens to find what changed

Evidence first. Usage fields and the Usage and Cost API show which tokens grew.

THE SIGNAL

'Nobody has looked at what changed' is the signal. The first step is to look.

> A property-maintenance app lets Claude update work orders through a tool. Two rules must always hold: Claude must never delete a work order, and every update written to the database must match the work-order schema.
>
> The team wants controls that hold even when the model makes a mistake.

Which two controls meet that bar? Choose two.

- A A system prompt line, in capitals, saying never to delete work orders
- B A hook that runs before each tool call and blocks any delete
- C Asking Claude to re-read each update and confirm it matches the schema
- D Validating each update against the schema in code before it is written
- E Lowering the temperature so the model's tool calls are more predictable

◆ PAUSE AND DECIDE

Which two controls meet that bar? Choose two.

A. A system prompt line, in capitals, saying never to delete work orders

Guidance, not a guarantee. A prompt asks; it cannot stop a call.

B. A hook that runs before each tool call and blocks any delete

Deterministic. The hook sees the call and blocks it before it runs.

C. Asking Claude to re-read each update and confirm it matches the schema

Still the model checking itself. Useful, but not a control.

D. Validating each update against the schema in code before it is written

Deterministic. Bad data stops in code, whatever the model produced.

E. Lowering the temperature so the model's tool calls are more predictable

Wrong layer, and current models reject non-default sampling settings.

THE SIGNAL

'Hold even when the model makes a mistake' means the control must live outside the model.

A two-person team at a recruitment start-up is adding a Claude feature to its Python back end. One developer proposes writing their own HTTP client, with hand-built retry logic for rate limits and a parser for the streaming events.

The team has no requirement that the official SDK cannot meet.

What should the team do?

A Use the official Python SDK, set its retry and timeout options, and use its streaming helper

B Write the custom client as proposed, so the team controls every detail of each request

C Adopt a third-party multi-provider framework in case another model is needed later

D Call the API without retries and show users an error whenever a request fails

◆ PAUSE AND DECIDE

What should the team do?

A Use the official Python SDK, set its retry and timeout options, and use its streaming helper

The SDK already retries rate limits and server errors with backoff, and parses streams.

B Write the custom client as proposed, so the team controls every detail of each request

Rebuilds what the SDK provides, and adds code two people must maintain.

C Adopt a third-party multi-provider framework in case another model is needed later

A new dependency for a need nobody has stated.

D Call the API without retries and show users an error whenever a request fails

Fails users on errors the SDK would have retried.

THE SIGNAL

'No requirement the SDK cannot meet' is the signal. Use what the platform gives you.

Module 1 in five lines

- **The exam tests decisions** a developer makes to ship Claude features that survive production.
- **53 questions, 120 minutes, pass at 720.** Domain 2 is a third of the marks.
- **Know the map:** models, Messages API, tools, MCP, Skills, Agent SDK, Managed Agents, Claude Code, platforms.
- **Ten rules decide most answers,** starting with: code enforces, prompts guide.
- **Copperlet** carries the examples from here on.

> COMING UP
>
> Next: how Claude actually generates text, and why the same prompt can give two different answers.